

GÖDEL'S SECOND INCOMPLETENESS THEOREM FOR GENERAL RECURSIVE ARITHMETIC

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Introduction

The Conditional Goldbach Theorem of H. A. POGORZELSKI [2] assumes the following metamathematical hypothesis:

(H2) *If equational (general) recursive calculus $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ is consistent, then the wff in $\omega(\mathcal{E}_{\mathbf{E}}^{\text{rec}})$ expressing this consistency of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ is not a theorem of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$.*

In this paper we prove that (H2) is true. Our method of proving (H2) is a modification of the method used by ROSE [4, 5]. The method used can be modified to provide proofs of GÖDEL's second incompleteness theorem for quantifier-free recursive arithmetics in general.

1. Gödel numbering of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$

We assume familiarity with our earlier paper [6]. In particular, we assume the formulation of the general recursive equational calculus, denoted by $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$, given in [6, § 3]. As in [6] we use PR as an abbreviation for "primitive recursive". The implication sign $\rightarrow$ is defined as in [1, Ch. 3]. We define $x \neq y$ by $1 \div |x, y| = 0$ (see [1, Ch. 3]). We assume familiarity with the PR functions and predicates defined in [3, pp. 69–71, 90–100] and [6, § 1]. We also assume familiarity with the PR functions and predicates concerning the Gödel numbering of the first-order PR arithmetic RA defined in [3, pp. 100–110] (with the corrections noted in [6, § 1]).

We assign numbers to the primitive symbols of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ as follows: We assign the number 17 to the atomic general recursive function letter F, and we assign numbers to the remaining primitive symbols of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ as in [3, p. 100]. We assign Gödel numbers to formulas and sequences of formulas of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ as in [3, p. 101]. Following ROBBIN [3, p. 101], for any formula A of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$, we let " A " denote the (formula) Gödel number of A .

Consider the PR functions and predicates with the following properties:

- i) The PR predicate $\text{var}(x)$ holds iff x is the Gödel number of an individual variable.
- ii) The PR function " k_n " gives the Gödel number of the numeral corresponding to the natural number n .
- iii) For each $n \geq 1$, the PR function $\eta(n, i)$ enumerates the Gödel numbers of n -place $\mathcal{E}$ -function constants.

¹⁾ The author wishes to acknowledge his gratitude to Professor H. A. POGORZELSKI for his generous help and encouragement.

- iv) The PR predicate $\mathcal{E}\text{-term}(x)$ holds iff x is the Gödel number of an $\mathcal{E}$ -term.
- v) The PR functions $\theta_1(x)$ and $\theta_2(x)$ are defined so that if $\mathbf{t}_1 = \mathbf{t}_2$ is an $\mathcal{E}$ -equation, then $\theta_1("t_1 = t_2") = "t_1"$ and $\theta_2("t_1 = t_2") = "t_2"$.
- vi) The PR function $Sub(x, t, e)$ must be defined so that if x , t , and e are the Gödel numbers of an individual variable x , an $\mathcal{E}$ -term t , and an $\mathcal{E}$ -equation E , respectively, then $Sub(x, t, e) = "S^x_t E"$.
- vii) The PR function $sub(x, n, e)$ must be defined by $sub(x, n, e) = Sub(x, "k_n", e)$.
- viii) The PR function $Ax(n)$ enumerates the Gödel numbers of the axioms of $\mathcal{E}^{\text{rec}}_{\mathbf{E}}$.

We now define a PR function $th(y)$ which enumerates in a certain order the Gödel numbers of the theorems of $\mathcal{E}^{\text{rec}}_{\mathbf{E}}$. Since, when using inference rules (SE) and (U), introduced in [6], we may infer theorems from arbitrarily many previous theorems of $\mathcal{E}^{\text{rec}}_{\mathbf{E}}$, we define below $th(y)$ in terms of prime factorization rather than the pairing function used by ROSE in [4]. We define $th(y)$ by:

$$\begin{aligned}
 th(5y) &= Ax(y); \\
 th(5y + 1) &= \begin{cases} \theta_2(th((y)_1)) * "=" * \theta_2(th((y)_2)) & \text{if } \theta_1(th((y)_1)) = \theta_1(th((y)_2)) \\ "0 = 0" & \text{otherwise;} \end{cases} \\
 th(5y + 2) &= \eta((y)_1 + 1, (y)_2) * "(" * \prod_{k \leq (y)_1} \theta_1(th((y)_{k+3})) * "(" = " * \eta((y)_1 + 1, (y)_2) * "(" * \\
 &\quad \prod_{k \leq (y)_1} \theta_2(th((y)_{k+3})) * "(""; \\
 th(5y + 3) &= \begin{cases} Sub((y)_1, (y)_2, th((y)_3)) & \text{if } var((y)_1) \text{ and } \mathcal{E}\text{-term}((y)_2) \\ "0 = 0" & \text{otherwise;} \end{cases} \\
 th(5y + 4) &= \begin{cases} ((y)_3)_1 & \text{if } P(y) \\ "0 = 0" & \text{otherwise,} \end{cases}
 \end{aligned}$$

where $P(y)$ denotes the PR predicate which holds iff $var((y)_1)$ and the sequence $th(((y)_2)_1), \dots, th(((y)_2)_{L((y)_2)}), ((y)_3)_1, \dots, ((y)_3)_{L((y)_3)}$ is the sequence of Gödel numbers of the equations (in the proper order) of an application of the uniqueness rule, (U), with

$$th(((y)_2)_{L((y)_2)}) = Sub((y)_1, "0", ((y)_3)_1)$$

and

$$((y)_3)_{L((y)_3)} = Sub((y)_1, "S" * (y)_1 * "'", ((y)_3)_1).$$

2. Gödel's incompleteness theorems for $\mathcal{E}^{\text{rec}}_{\mathbf{E}}$

The proofs of the lemmas and Main Theorem of this section parallel the proofs of the corresponding results in ROSE [4, 5].

Lemma 1. *If $E_1, \dots, E_n \vdash E_{n+1}$ in $\mathcal{E}^{\text{rec}}_{\mathbf{E}}$, then there exists a PR function $V(X_n)$, whose construction depends on the derivation, such that*

$$\vdash_{\mathcal{E}} th(x_1) = "E_1" \rightarrow [\dots \rightarrow [th(x_n) = "E_n" \rightarrow th(V(X_n)) = "E_{n+1}"] \dots].$$

The PR function $\varepsilon(x)$ must be defined so that if $\mathbf{t}_1 = \mathbf{t}_2$ is an $\mathcal{E}$ -equation, then $\varepsilon("t_1 = t_2") = "t_1 \neq t_2"$. Next, for any $n \geq 1$ and any n -place PR function constant f , $\{f(X_n) = x_0\}$, a PR function of $x_1, \dots, x_n, x_0$, must be defined so that

$$\{f(X_n) = x_0\} = "f(k_{x_1}, \dots, k_{x_n}) = k_{x_0}"$$

Lemma 2. *For any $n \geq 1$ and any n -place PR function constant f , there exists an $(n. + 1)$ -place PR function $p_f(X_n, x_0)$ such that*

$$\vdash_{\mathcal{E}} f(X_n) = x_0 \rightarrow th(p_f(X_n, x_0)) = \{f(X_n) = x_0\}.$$

Lemma 3 (GÖDEL's first incompleteness theorem). *Let*

$$N = \text{"sub("}x_0\text{", }x_0, x_0) \neq th(x_1)\text{"}.$$

Then

$$\vdash_{\mathcal{E}} \text{sub("}x_0\text{", }k_N, k_N) = \text{"sub("}x_0\text{", }k_N, k_N) \neq th(x_1)\text{"},$$

and, if $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ is consistent, then the equation

$$\text{sub("}x_0\text{", }k_N, k_N) \neq th(x_1)$$

is true but not provable in $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$.

Main Theorem (GÖDEL's second incompleteness theorem). *If $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$ is consistent, then the equation $th(x_0) \neq \varepsilon(th(x_1))$, that is, the equation asserting the consistency of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$, is not a theorem of $\mathcal{E}_{\mathbf{E}}^{\text{rec}}$.*

References

- [1] GOODSTEIN, R. L., Recursive number theory. North-Holland, Amsterdam 1964.
- [2] POGORZELSKI, H. A., On the Goldbach conjecture and the consistency of general recursive arithmetic. J. reine angew. Math. **268/269** (1974), 1–16.
- [3] ROBBIN, J. W., Mathematical logic: a first course. W. A. Benjamin, New York 1969.
- [4] ROSE, H. E., On the consistency and undecidability of recursive arithmetic. This Zeitschr. **7** (1961), 124–135.
- [5] ROSE, H. E., Some metamathematical results in recursive arithmetic. This Zeitschr. **13** (1967), 381–384.
- [6] RYAN, W. J., The equivalence of equational and sentential general recursive arithmetics. J. London Math. Soc. (2), **14** (1976), 463–475.

(Eingegangen am 8. Januar 1977)